MTH 346/646 homework cover sheet

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What topics did this assignment cover? How comfortable do you feel with each of those topics?

Lutz-Nagell, it seems. I don't feel comfortable with the material at this point, so I'm "taking the L" on this homework, as they say, to give myself some time to try to go back and catch up since I feel like I've let myself slip through the cracks and don't really know what's going on at this point :(

Hoping next homework will go better!

In the course of doing this assignment, where did you get stuck? How did you get unstuck?

I attempted problem 1 and started writing some Magma code, but I realized I must be misunderstanding what $C(3^k)$ meant. I thought it meant perhaps that nP was congruent to $(0 : 1 : 0)$ mod 3^k, but that must not be right.

Is there anything you've done here that you're not confident about, that you want me to take a really close look at, or that you want me to suggest an alternate approach to?

N/A

1. Let $C : y^2 = x^3 + 8x + 1$. Then $C(\mathbb{Q}) \cong \mathbb{Z}$ and a generator is $P = (2, 5)$.
For each k, $1 \le k \le 8$ determine the smallest positive integer n so that $nP \in C(3^k)$.

Proof. I used this Magma code to attempt it... but I'm not sure I have the right idea of what $C(3^k)$ is since this gave no results for $n \le 500$.

```
C := EllipticCurve([8, 1]);
P := C![2, 5];

for k in [1..8] do;
  printf "\n----------\nk=%o\n----------\n", k;
  modulus := 3^k;

  for i in [1..500] do;
    iP := i * P;
    // check if iP congruent to (0 : 1 : 0) mod 3^k
    if IsIntegral(iP[1]/modulus) and
       IsIntegral((iP[2] - 1)/modulus) and
       IsIntegral(iP[3]/modulus) then;
     printf "i=%o : %o\n", i, iP;
    end if;
  end for;
end for;
```

□

2. Let $C : y^2 = x^3 + ax^2 + bx + c$ and let p be a prime for which $C/\mathbb{F}_p$ is non-singular mod p. Show that the reduction homomorphism $\phi_p : C(\mathbb{Q}) \to C(\mathbb{F}_p)$ (defined in the day 9 notes) has kernel equal to $C(p)$.

3. $*$ Use the Lutz-Nagell theorem and problem 2 above to show that if $C : y^2 = x^3 + ax^2 + bx + c$ is an elliptic curve with torsion subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, then $C/\mathbb{F}_p$ must be singular for $p = 2$, $p = 3$, $p = 5$, and $p = 7$.